The Complete Injection–Suction and First-Cusp Atlas for Quadratic Polubarinova–Galin Flow

HCS-C368 / HEN-O352

4 September 2026

Abstract

For the normalized quadratic map $f(\zeta,t) = a(t)\zeta + b(t)\zeta^2$ we solve the constant-rate Polubarinova–Galin equation on its entire smooth univalent branch. The complex coefficient a^2b is conserved, normalized area is affine in physical time, injection is global and asymptotically circular, while noncircular suction ends at an explicitly timed ordinary semicubical cusp. Circular collapse and nonsmooth or invalid initial faces are separated. This round is the **coefficient-reduction owner**.

1 Normalized flow and exact reduction

Let $a(t) > 0$, $b(t) \in \mathbb{C}$, and

$$f(\zeta,t) = a(t)\zeta + b(t)\zeta^2, \qquad \operatorname{Re}\big[f_t\,\overline{\zeta f_\zeta}\big] = q \quad (|\zeta| = 1), \tag{1}$$

where $q \in \mathbb{R}$ is constant. The normalization $f(0,t) = 0$ and $f_\zeta(0,t) = a(t) > 0$ fixes translation and rotation. A *smooth univalent solution* means $a > 2|b|$; then f is conformal on a neighborhood of $\overline{\mathbb{D}}$ and parametrizes a smooth Jordan boundary.

Theorem 1 (Complete smooth coefficient branch)**.** *For smooth univalent initial data, equation* (1) *is equivalent to*

$$\dot a = \frac{aq}{a^2 - 4|b|^2}, \qquad \dot b = -\frac{2qb}{a^2 - 4|b|^2}. \tag{2}$$

On its unique maximal smooth interval,

$$\kappa := a^2 b \quad \textit{is constant}, \qquad M_0 := a^2 + 2|b|^2 = \frac{\operatorname{Area}(f(\mathbb{D}))}{\pi}, \qquad \dot M_0 = 2q. \tag{3}$$

Put $u = a^2$ and $u_c = (4|\kappa|^2)^{1/3}$, taking $u_c = 0$ when $\kappa = 0$. Then

$$M_0 = F(u) := u + \frac{2|\kappa|^2}{u^2}, \qquad F'(u) = 1 - \frac{4|\kappa|^2}{u^3}, \tag{4}$$

and smooth univalence is exactly $u > u_c$.

If $q > 0$, the solution is smooth for every $t \ge 0$, u increases to infinity, and $2|b|/a \to 0$. If $q = 0$, the map is stationary.

Proof. On $|\zeta| = 1$, expanding the left side of (1) and equating its constant and first Fourier modes gives

$$a\dot a + 2\operatorname{Re}(\dot b\,\bar b) = q, \qquad a\dot b + 2\dot a b = 0. \tag{5}$$

The determinant is $a^2 - 4|b|^2$, so (2) follows. The second identity gives $(a^2b)^{\cdot} = 0$, and both identities give $\dot M_0 = 2q$. The area formula follows either from the coefficient area theorem or from $\frac{1}{2}\int_0^{2\pi} \operatorname{Im}(\bar f f_\theta)\,d\theta$. Substituting $b = \kappa/u$ proves (4).

Indeed,

$$f(z_1) - f(z_2) = (z_1 - z_2)\{a + b(z_1 + z_2)\}, \qquad f'(\zeta) = a + 2b\zeta.$$

Thus $a > 2|b|$ gives injectivity and nonvanishing derivative on $\overline{\mathbb{D}}$; equality puts the sole critical point on its boundary, and $a < 2|b|$ puts it inside. Since F is strictly increasing on (u_c, ∞), $M_0(t) = M_0(0) + 2qt$ uniquely determines that branch. For $q > 0$ it exists globally with $u \to \infty$ and $2|b|/a = 2|\kappa|/u^{3/2} \to 0$. For $q = 0$, (2) vanishes. ∎